

CHE 205, Fall 2019, Problem Set 6

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Due date: 11:59pm, Fri, 10/25/2019

Submit two files, one for each problem, on blackboard.

Problem 1 (8 pts). Re-implement Example 5.7 of the textbook using **backward Euler** method.

Problem 2 (8 pts). Re-implement Example 5.7 of the textbook using **second-order RK** method.

Example 5.7: A VBA Program for the Euler Method

In certain circumstances, it can be more efficient to write a VBA program to solve ODE-IVPs numerically. Such cases include times when it is convenient to simply alter a subroutine subprogram that defines the right-hand side function and also when there is a large system of ODE-IVPs involved. This example shows the development of a VBA program to implement Euler's method for any number of simultaneous ODEs. First, a program Specification is given that defines the Excel interface to the program followed by an Algorithm Design, Coding in VBA, and Testing using a problem previously solved using only Excel.

SPECIFICATION

The following “mock-up” of an Excel spreadsheet for solving any number of ODE-IVPs using the Euler method represents a program specification:

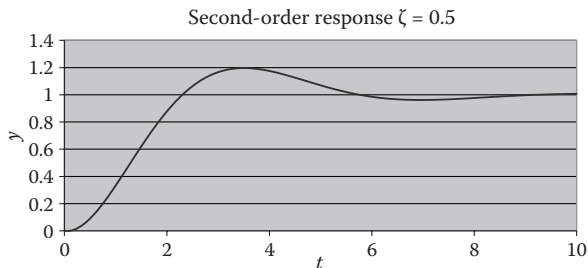


FIGURE 5.8 Response of a second-order system to a step response.

	A	B	C	D	E	F
1	2	N	0.01	h	100	TimeMax
2	Time	y1	y2	y3	...	yN
3	0	Initial Conditions Go Here				
4	0.01	y11	y21	y31	...	yN1
5	0.02	y12	y22	y32	...	yN2
6	⋮	⋮	⋮	⋮	⋮	⋮
7	TimeMax	y1N	y2N	y3N	...	yNN
8						
9	Cell A1 contains the number of simultaneous ODE IVPs to solve. Cell C1 holds the time step and cell E1 the maximum time required. The second row contains appropriate labels (generic ones are shown here). Cells labeled y11, y21, etc. contain values produced by the program. The program "reads" N, h and MaxTime as well as the initial conditions (cells B3 ...). The program computes, one row at a time, the values of y and outputs them to the appropriate row of the spreadsheet.					
10						
11						
12						
13						
14						
15						
16						
17						

ALGORITHM DESIGN

The structure chart of Figure 5.9 shows an algorithm design for the main program and for a subroutine to calculate the right-hand side functions. A dictionary of variables is also given.